Electronics has the highest return rate after quality issues are removed

Operations and data-quality report | Synthetic order and return data

Executive Summary

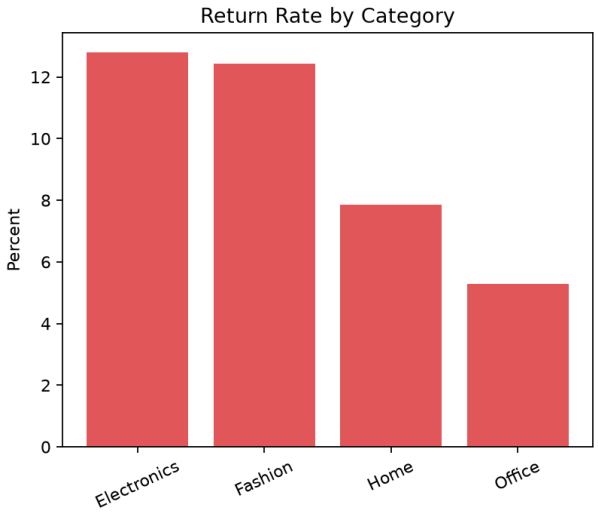
701	1	3	12.8%
RAW ROWS	DUPLICATE ORDER	MISSING PRODUCT IDS	ELECTRONICS RETURN RATE

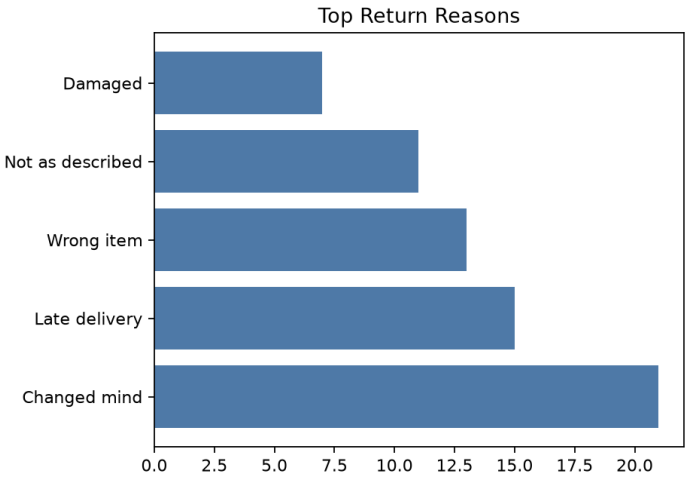
- The quality audit found one duplicate order and 3 missing product IDs; these records were removed before calculating return rates.
- Electronics has the highest category return rate at 12.8%, compared with 5.3% for Office.
- Changed mind is the most frequently recorded return reason with 21 cases.

Why the cleaning step matters

Duplicate orders inflate the denominator or return count, while missing product IDs prevent reliable product-level follow-up. The project records these issues before removing them so the analysis remains auditable.

Electronics returns stand out as the main operational issue





Category	Orders	Returns	Return rate	Avg delivery days
Electronics	172	22	12.8%	4.1
Fashion	177	22	12.4%	4.3
Home	178	14	7.9%	4.1
Office	170	9	5.3%	4.2

Recommendations

- Review sizing, product descriptions and supplier quality for Electronics, then monitor return rate by SKU.
- Investigate changed mind cases using fulfilment and quality-control records rather than relying on the reason label alone.
- Add validation rules that require product ID for every order and prevent duplicate order IDs at ingestion.

Method and limitations

Return rate equals returned orders divided by unique cleaned orders within each category. Records with duplicate order IDs or missing product IDs are excluded. The sample is synthetic; reason labels describe association, not confirmed root cause.

Reproducibility

Source: data/orders_and_returns.csv. Analysis: analysis.py. Generated outputs: data_quality_report.csv, return_rate_by_category.csv, return_reasons.csv and returns_analysis.png.